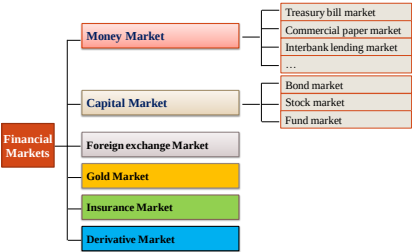


Introduction to Chinese Financial System

LECTURE 5 MONEY MARKET IN CHINA

INTRODUCTION OF MONEY MARKET

Financial markets



What is a money market

Characteristics of money market instruments

Money market classification

- 1 Interbank Lending Market (银行间同业拆借市场)
- 2 Short-term Government Bond Market (短期国债市场)
- 3 Market Negotiable Instruments (票据市场)
- 4 Market for Negotiable Certificates of Deposit (同业存单市场)
- 5 Market for Repurchase Agreements (回购协议市场)

INTERBANK LENDING MARKET
(银行间同业拆借市场)

Definition of interbank lending market

- The interbank lending market is a market for short-term (mostly overnight) credit lending between financial institutions to adjust temporary positions and meet liquidity needs.
- Broadly speaking, the interbank lending market refers to the market where financial institutions carry out short-term financing.
 - The financing is not only limited to making up or adjusting the capital position, but also becomes a market for financial institutions, especially commercial banks, to make up for the lack of liquidity and effectively use funds to reduce idle funds.

Cause of interbank lending market

- Cause: the requirement of central bank for the statutory **deposit reserve of commercial banks**.

Origin of interbank lending market

- The interbank lending market first appeared in the United States, and the United States was the first in the world to legally require commercial banks to pay deposit reserves to the central bank.

Development of interbank lending market

- Original function: to meet the needs of financial institutions in the daily business activities of surplus and deficit adjustment.
- Development: securities firms have also joined the interbank lending market, and the transaction object is no longer limited to the deposit reserve of commercial banks.

Characters of interbank lending market

Participants of interbank lending market

- Generally, those who can enter the market must be financial institutions.

The way of inter-bank lending

Development of China's Interbank Market

The interest rate in the interbank market

- The interbank offered rate refers to the interest rate at which banks lend and borrow short-term funds between each other, including the “ask rate” and the “bid rate”.
 - Bid Rate: the interest rate at which banks are willing to borrow;
 - Ask Rate or Offered Rate: the interest rate at which banks are willing to lend.
- For the same bank, its Bid Rate is always less than the its Offered Rate, and the difference is the bank's profit.
- The Interbank Offered Rate is one of the benchmark interest rates in financial markets.

LIBOR (London Interbank Offered Rate)

- The interest rate that is heavily used in Interbank lending is LIBOR (London Interbank Offered Rate).
- LIBOR is an interest rate average calculated from estimates submitted by the leading banks in London. It is the rate at which a leading bank in London is willing to offer funds.
- LIBOR is now used as the basis for most floating rates in international financial markets. It is used as the funding cost for banks to raise money from the market for lending.
- China's external financing cost is generally based on the LIBOR rate plus a certain percentage point.

Goodbye LIBOR, hello SOFR

SHIBOR

- Shanghai Interbank Offered Rate (SHIBOR).
- Referring to the experience of London Interbank offered Rate (LIBOR), China officially launched SHIBOR in January 2007.
 - SHIBOR is calculated from rates quoted by 18 banks, eliminating the four highest and the four lowest rates, and then averaging the remaining, including 8 maturity varieties from overnight to 1 year.
 - SHIBOR is pure credit quoted interest rate, no matter whether there is a transaction.
 - SHIBOR has been expected to become the Chinese version of LIBOR since its birth. After more than ten years of cultivation, SHIBOR has gradually become the benchmark interest rate recognized by the market.

THE TREASURY BILL MARKET
(短期国债市场)

T-Bills (Treasury Bills)

- T-bills, also known as "Treasury bills", are the shortest maturity bonds issued by the government. Its maturity is generally 3 months, 6 months, 9 months and 12 months.

The function of Treasury bills

Characteristics of Treasury bill

The ways of issuing China's T-bills

The issuance of Treasury bills in China

The circulation market of Treasury bonds

- The secondary market for Treasury bills is the trading market for the circulation and transfer of issued Treasury bills. Generally, it is a transaction between underwriters, subscribers, holders, the government.
- Functions:
 - Reduce liquidity risk for holders.
 - Continuous trading to form the market price and market interest rate.
 - Provides better investment opportunities for potential investors.
 - An important place for the central bank to conduct open market operations.

The circulation market of Treasury bonds

- Trading market is divided into two types:
 - Stock exchange trading, refers to trading conducted in the exchanges;
 - Over-the-counter trading, which is not conducted on the floor of an exchange.

Investors of Treasury bills

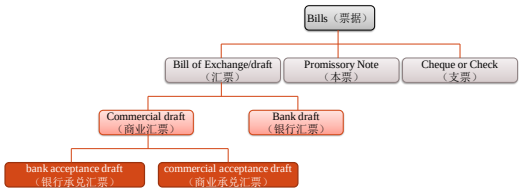
The history of China's Treasury bill market

MARKET FOR NEGOTIABLE INSTRUMENTS
(票据市场)

What are negotiable instruments?

Main types of bills

Main types of bills



Functions of bills

The bill market

- The bill market is the place where bills are circulated and transferred.
- The bill market is divided into:
 - Bill acceptance market (issue market):dealing with bill acceptance business;
 - Bill discount market:the place where bills are directly discounted.
- The bill market is a place for acceptance and discount services between the drawer and the receiver.

Classification of bill market

- The bill market in China mainly includes:
 - The market for Bank Acceptance Drafts
 - The market for Commercial Acceptance Drafts
 - The market for Commercial Papers

Bank Acceptance Drafts

- A bank acceptance draft is a bill drawn by the payee or payer in a commodity transaction and accepted by the bank with its signature and seal on the bill, agreeing to pay on the due date of the bill.
- Bank acceptance draft was originally produced in international trade financing, and has been widely used in the ordinary trade settlement business between enterprises.

Commercial Acceptance Drafts

- A commercial acceptance draft is a bill drawn by the payee and accepted by the payer, or drawn and accepted by the payer. The acceptor (payer) is a general industrial and commercial enterprise, not a bank.
- The drawer must be a legal entity that opens an account in a commercial bank, should be based on legal commodity transaction, and after the draft is accepted, the acceptor (namely payer) has the responsibility to pay the draft unconditionally when due.
- At the same time, the draft can be discounted to the bank, also can be circulated and transferred.

Differences between commercial and bank acceptance draft

Commercial Paper

- Commercial Paper ("CP") is a short-term financing instrument issued directly to money market investors by reputable companies for the purpose of raising capital and for the purpose of short-time financing.
 - Most of them are promissory notes with the drawer as the payer. The drawer promises to pay a certain amount to the payee at a certain time and place.
 - It is an unsecured debt certificate of the drawer.

Commercial Paper

- Commercial Paper originated in the United States in the 1920s.
 - American commercial paper is a promissory note.
 - British commercial paper is a bill of exchange.

Issuers and investors of commercial paper

- Issuers
 - Large companies with strong capital and excellent reputation, financial companies of large enterprises, financial companies, bank holding companies and public utilities, etc.
 - Smaller or lesser-known companies with low credit ratings must be backed by credit from companies with higher credit ratings (called credit backed paper) or secured by high quality assets (called mortgage backed commercial paper).
- Investors
 - Institutional investors, including pension funds, money market mutual funds, non-financial corporations, commercial banks, insurance companies, investment companies and others.
 - Individual investors only account for a small proportion.

Issuance of commercial paper

How is a commercial paper issued?

Trading of commercial paper

- The secondary market for commercial paper is over-the-counter block trading.
 - In the U.S. commercial paper secondary market, the minimum trade size is \$100,000. Small investors can only participate indirectly in commercial paper by investing in money market mutual funds.
 - In China, the minimum trade size is ¥1,000,000 (1 million RMB)
- The secondary market trading is less active around the world.
 - The bills have a short maturity and most of them are held to maturity;
 - The bills are highly heterogeneous, with different maturities, denominations and interest rates, making them difficult to trade.

Commercial paper and commercial acceptance

Commercial paper in China

- For some reasons, in China, we have a different name for commercial paper, which is “short-term financing bill”.
- China's short-term financing bills started in 1988, but were suspended in 1997 because some enterprises could not repay on schedule.
- They were resumed in 2005.
- In essence, the “short-term financing bills” reissued by China in 2005 are commercial papers.

Short-term financing bills

- Short-term financing bills refer to securities issued by non-financial enterprises in the inter-bank bond market, agreed to repay the principal and interest within a certain period of time.
- Short-term financing bills in China have similar characteristics of commercial paper in the U.S. market, except that the maturity of short-term financing bills in China could reach one year and banks cannot issue them.

Why issue short-term financing bills?

History of short-term financing bills in China

New characteristics of short-term financing bills

Development of China's bill market

- Commercial acceptances (commercial acceptance drafts) and commercial papers (short-term financing bills and ultra-short-term financing bills) are the main financial instruments China's bill market.

Participants of Bill Market

Development of China's bill market

- According to data released by the People's Bank of China in 2023,
 - In 2023, the amount of commercial acceptances and discount was 31.3 trillion yuan and 23.8 trillion yuan, respectively.
 - By the end of 2023, the balance of commercial acceptances was 18.6 trillion yuan, down 2.7% year on year; The discount balance was 13.3 trillion yuan, up 2.1% year on year.

Major problems in China's bill market

MARKET FOR CERTIFICATES OF DEPOSIT
(同业存单市场)

Market for certificates of deposit

- Inter-bank Certificates of Deposit Market
- Negotiable Certificates of Deposit (CDs) Market

Negotiable Certificates of Deposit (CD)

- Negotiable certificates of deposit are fixed-denomination deposit certificates issued by banks that can be transferred and circulated.
 - The main difference between them and ordinary time deposits lies in their transferability, which enables investors to transfer the certificates in the secondary market when they need funds.
- Market Functions and Participants:
 - Banks issue CDs to absorb funds, while investors such as enterprises and individuals purchase them in order to obtain higher returns than ordinary deposits and maintain a certain level of liquidity.

Inter-bank Certificate of Deposit

- Inter-bank certificates of deposit are book-entry time deposit certificates issued by banking deposit-taking financial institutions in the national inter-bank market. They are a financial instrument for fund intermediation among financial institutions and are mainly used to regulate the liquidity of funds among banks.
- Issuers and Traders:
 - The issuers are banking deposit-taking financial institutions, including policy banks, commercial banks, etc. The purchasers are also financial institutions.
 - For example, small banks can raise funds from large banks or other financial institutions by issuing inter-bank certificates of deposit to meet their own capital needs or optimize their asset-liability structures.

Background of CDs: Regulation Q

- After the Great Depression in 1929, the Federal Reserve Board of U.S.A issued a series of financial regulations, and arranged them in alphabetical order from item A to Z, among which the regulation of deposit interest rate is item Q. Therefore, this regulation is called "Regulation Q".

Creation of negotiable certificates of deposit

Functions of NCDs

Market Transactions of NCD

- Primary Market:
- Secondary Market:

Markets for CDs in China

REPURCHASE AGREEMENT MARKET
(回购协议市场)

Repurchase agreements

- Repurchase Agreement (Repo): an agreement whereby a securities seller agrees to repurchase the securities at an agreed price after a certain period of time.
- Reverse Repurchase Agreement (Reverse Repo) : an agreement in which a security is purchased with the agreement to resell the purchased securities at an agreed price at a future date.
- A repurchase agreement is essentially a loan pledged, and the collateral is the security being traded.

Characteristics of the repurchase agreements

Functions of repurchase agreements

Classification of repurchase agreements

- Pledged Repo(质押式回购)
 - Pledged repo, or "mortgage repo", is a short-term financing business in which the two parties pledge their rights with bonds.
 - Bond pledged repo means that the bond holder (the repo party) pledges the bond to the financing party and agrees with the buyer to buy back the bond from the financing party at an agreed price on a certain date in the future.
 - In the process of a pledged repurchase transaction, ownership is not transferred, and the bonds are generally frozen and held in custody by a third-party custodian institution and unfrozen at maturity.
 - China launched the bond pledged repo in 1997.

Classification of repurchase agreements

- Outright Repo(买断式回购)
 - Outright repo, or "buy-out repo", refers to a transaction in which a bond holder (the repo party) sells a bond to a bond buyer (the reverse repo party) with the agreement that the seller will buy back the bond from the buyer at an agreed price at a future date.
 - In the process of an outright repurchase transaction, the ownership of bonds will be transferred, which also has a certain short selling function.
 - The introduction of outright repo changed the pattern that the pledged bonds are frozen in large quantities, thus effectively revitalizing the bond stock and activating the cash bond trading.
 - China launched the bond buy-out repo in 2004.

Repurchase agreement (Repo) market

- A repurchase agreement transaction consists of a repurchase agreement from one party and a reverse repurchase agreement from the other party.
- The repurchase agreement market refers to the market for short-term financing transactions through repurchase agreements.

Participants in Repo market

Operation forms of the Repo market

Proceeds from repurchase agreements

- Repo rate: interest rate that is stated on an annual basis using 360 days.
- Determinants of repurchase rate (Repo Rate):

Risks of repurchase agreements

China's repurchase agreement market

- Interbank repurchase agreement market
- Exchange repurchase agreement market

Summary

Characteristics of money markets

The role of money markets
